Material Selection for Wind Blade in Wind Turbine Design

Date: December 3rd, 2024
Name: Tom Dang
Course: Engineering Materials

Abstract

This study focuses on the selection of materials for wind turbine blades, aiming to balance mechanical performance, environmental sustainability, and cost efficiency. The analysis incorporates critical factors such as fracture toughness, bending stiffness, density, and fatigue resistance, alongside lifecycle energy consumption, carbon emissions, and recyclability. A penalty function method normalizes these parameters and ranks materials based on their performance. Medium carbon steel emerges as the optimal choice due to its excellent balance of strength, low environmental impact, and cost-effectiveness. Stainless steel is a versatile alternative, while nickel-based superalloys provide unmatched mechanical properties but suffer from high energy and carbon costs. The findings offer a structured framework for material selection tailored to the design priorities of wind turbine blades.

Table of content

Abstract **1**
Background **3**
Problem statement 3
Mechanical requirement 3
Steps for selecting materials **4**
Choosing through fracture analysis 4
Choosing through mechanical model 9
Eco audit **11**
Justification of material choice **14**
Create penalty function 14
Calculation of manufacturing cost 15
Analysis from different viewpoints 16
Conclusion **16**

Background

Problem statement

Aircraft engine consists of multiple blades rotating around a large disk, which is driven by the long shaft in the center line. The blades are rotated at very high to push the air backward past the engine. The blade design is simplified in this problem as a rectangular prism of $900 \times 300 \times 20\,mm$ with thickness of $6\,mm$. The typical complex airfoil shape is complex for the purpose of selecting material for mechanical design.

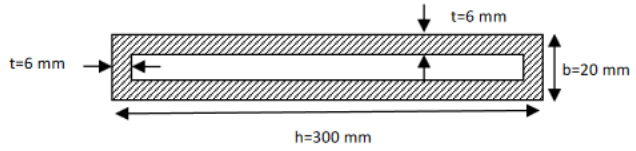


Figure 1: Simplified design of wind blade

The mass of the blades rotating at high speed not only creates high stresses in the blades themselves, it also requires that the fan disk be strong enough to hold the blades, further increasing the weight of the system. Since increased engine weight increases the takeoff weight and reduces the payload, it is desired to minimize the weight of the fan blade, subject to bending stiffness of the blades.
Cost is always a constraint in jet engines, particularly commercial ones, and it is desired to keep blade cost as low as possible. Cost is not only a function of starting material, but also fabrication. To work out the fabrication costs, calculate the surface area, internal (inside cavity) and well as external, and multiply by 6000 CAD/m^2. Blades are also experiencing a tensile stress (σ) caused by centrifugal loading. For a given angular velocity, assume that this stress scales just with the density ρ of the material, such as $\sigma(MPa) = 0.05\,\rho(\frac{kg}{m^3})$
Tolerance to damage (dents, cracks) from impact of foreign objects (rocks, birds) is also important. A 0.5 mm deep impact-induced crack should not propagate under the cyclic loads imposed by centrifugal force. Fast fracture will occur if the fracture toughness reaches the value of $\sigma\sqrt{\pi a}$, a is the value of impact-induced crack size in metres.
In addition to the centrifugal loads, blades are subjected to lateral vibrational loads due to air turbulence, system excitation, etc. The method used to reduce vibrational fatigue is to keep the natural frequency as high as possible. Natural frequencies of blades scale as $\sqrt{E/\rho}$, so the answer is to make this as large as possible. The temperature of the fan blade will reach a maximum of 150°C during operation.

Mechanical requirement

- To meet the air intake requirements the blades of the turbine must be of specific size and shape. It must have the dimensions of 900mm x 300mm x 20m for the exterior; but it must also have a hollow center so the thickness of the sides must be 6mm.

- Since increased engine weight increases the takeoff weight and reduces the payload, it is desired to minimize the weight of the fan blade, subject to bending stiffness of the blades. → **less density, more bending stiffness**

- Cost is always a constraint in jet engines, particularly commercial ones, and it is desired to keep blade cost as low as possible. Cost is not only a function of starting material, but also fabrication. → **As cheap as possible**

- By assuming that a material will last 100,000 cycles if the tensile stresses are held below one half of the ultimate tensile strength → Minimum fatigue cycle of **100,000**

- Blades are also experiencing a tensile stress (σ) caused by centrifugal loading. For a given angular velocity, assume that this stress scales just with the density ρ of the material → Constraint equation for selecting yield stress and density. $\boldsymbol{\sigma}$ (MPa) $\geq 0.05\boldsymbol{\rho}$ $(\frac{kg}{m^3})$

- Tolerance to damage (dents, cracks) from impact of foreign objects (rocks, birds) is also important. Maximum crack of 0.5 mm on the surface

- The method used to reduce vibrational fatigue is to keep the natural frequency as high as possible. Natural frequencies of blades scale as $(E/\rho)^{1/2}$, so the answer is to make this as large as possible (or make ρ/E as low as possible) → Maximize the equation of natural vibration
- Temperature of the fan blade reaches the maximum temperature of 150 degree Celsius => Maximum service temperature at least 150 C
- All the measurements are in the metric unit system.

Steps for selecting materials

Choosing through fracture analysis

Based on the provided data, the blade is assumed to be a rectangular prism with length (l), width (w), and depth(d) as $900 \times 300 \times 20\ mm$. The structure is hollow with $6\ mm$ of thickness (t). The total surface area on the outer surface is calculate as the sum of every product pair of length, width, and depth, which is $0.588\ m^2$. For the inner surface area, every dimension is subtracted by twice the thickness, the calculation of surface area is the same approach, which given to be $\approx\ 0.530\ m^2$.

From the mechanical requirements, there are important parameters

- The minimum maximum service temperature is 150℃
- Natural frequency ($\sqrt{E/\rho}$) to be maximized
- Fracture toughness $K_{IC} \geq\ \sigma\sqrt{\pi a}$, stress is proportional to density as $\sigma\ =\ 0.05\rho$, $a\ =\ 0.5mm\ =\ 0.5\ *\ 10^{-3}m\ =\ 5\ *\ 10^{-4}m$ and $\pi\ =\ 3.14$ (assumed)
- Tensile stress is less than one half the ultimate tensile strength $\sigma \leq\ \frac{\sigma_{uts}}{2}$, where $\sigma\ =\ 0.05\rho$

Assuming the material of interest is general so that the material in level 2 is good enough for the finding. The strategy is setting the limit of service temperature, then selecting material that has a ratio of fracture toughness and its inequality equation at least larger than 1. Then set the ratio of twice tensile stress and ultimate tensile strength less than 1. Then, ranking according to the natural frequency from largest to smallest

Starting with setting the limit of minimum maximum service temperature, the number of material passed this criteria is 55 out of 100 material. Then, find the material that pass the fracture, the equation implemented as $\frac{K_{IC}}{0.001981161276\times\rho}$ and use box selection to pass everything above 1. The material range is down to 28 out of 100 materials. The equation setted for fatigue cycle is $\frac{\sigma_{uts}}{0.1\rho}$ and use box selection to get everything below 1. The material is limited down to 21 out of 100.

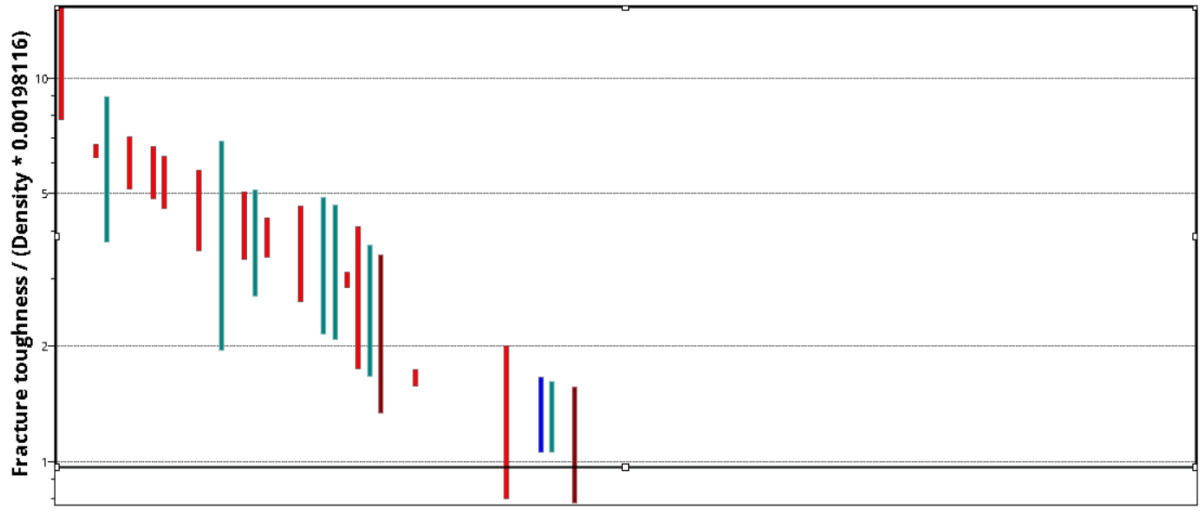


Figure 2: Selecting material based on fracture toughness and its inequalities equation

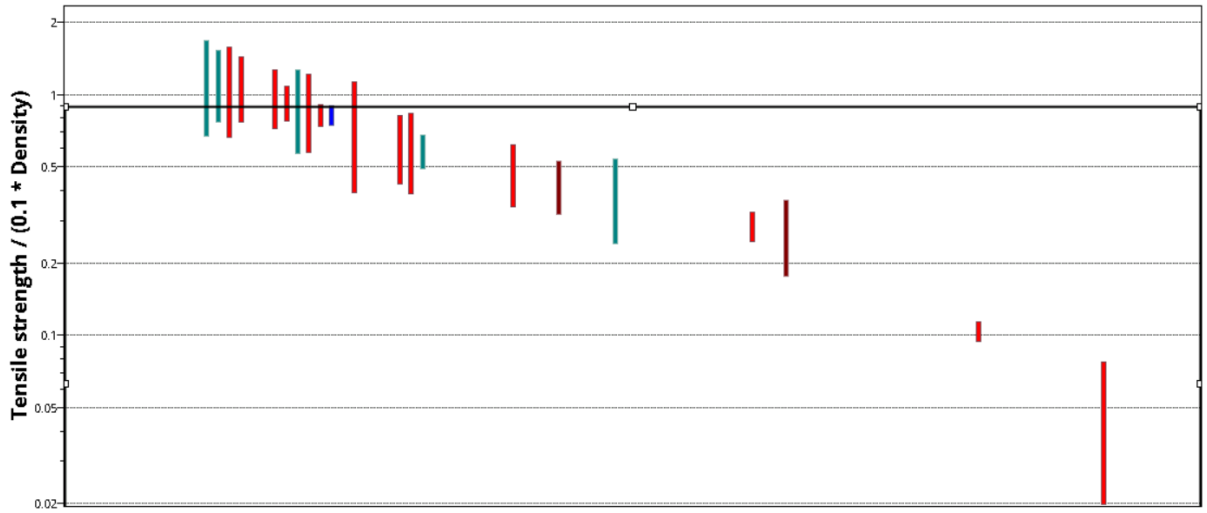


Figure 3: Selecting material based on the fatigue cycle equation

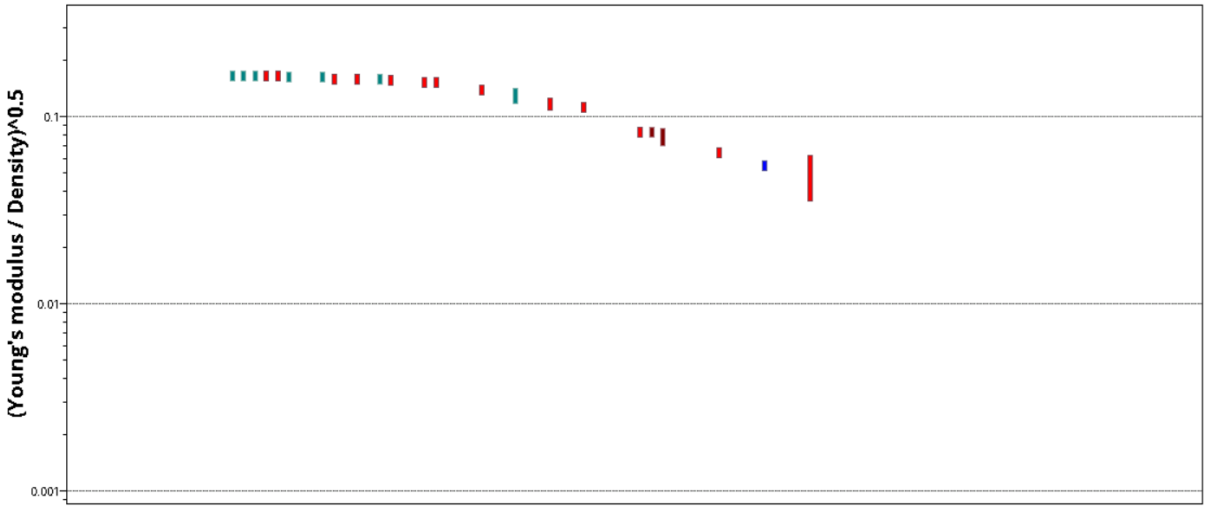


Figure 4: Final result of natural frequency for all passed material

In this observation, the reason for choosing material slightly less than 1 is because there are two materials that are mostly more than 1, it is reasonable not to choose the materials that are hard to engineer in such a fatigue cycle. The natural frequency is generally high and there are many metal options to choose from. Thus the variance in natural frequency of the top 11 material is very low, which makes price an important decision factor and so does density and stiffness.

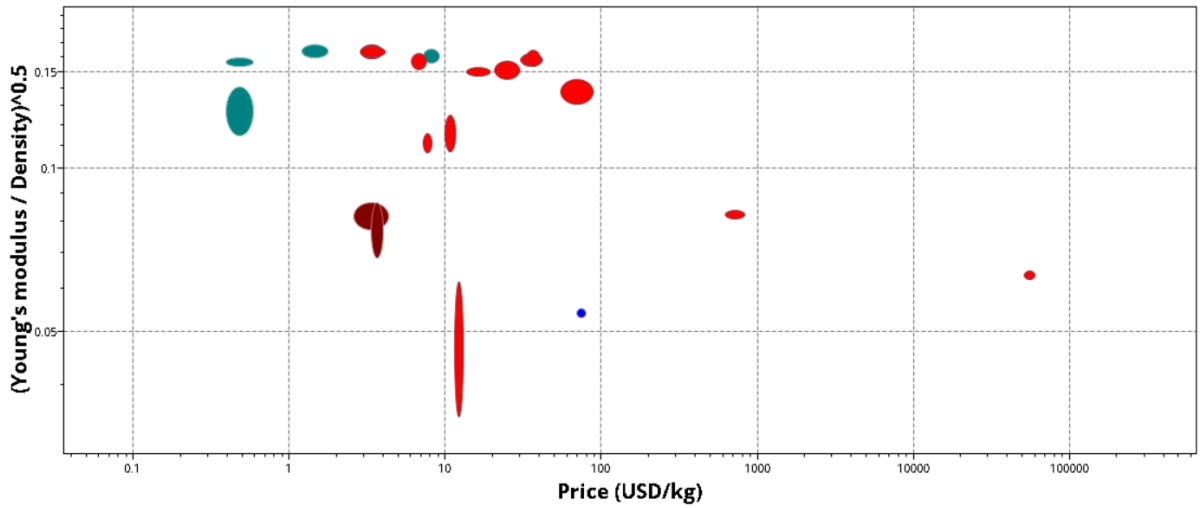


Figure 5: Material's natural frequency in comparison with price of material

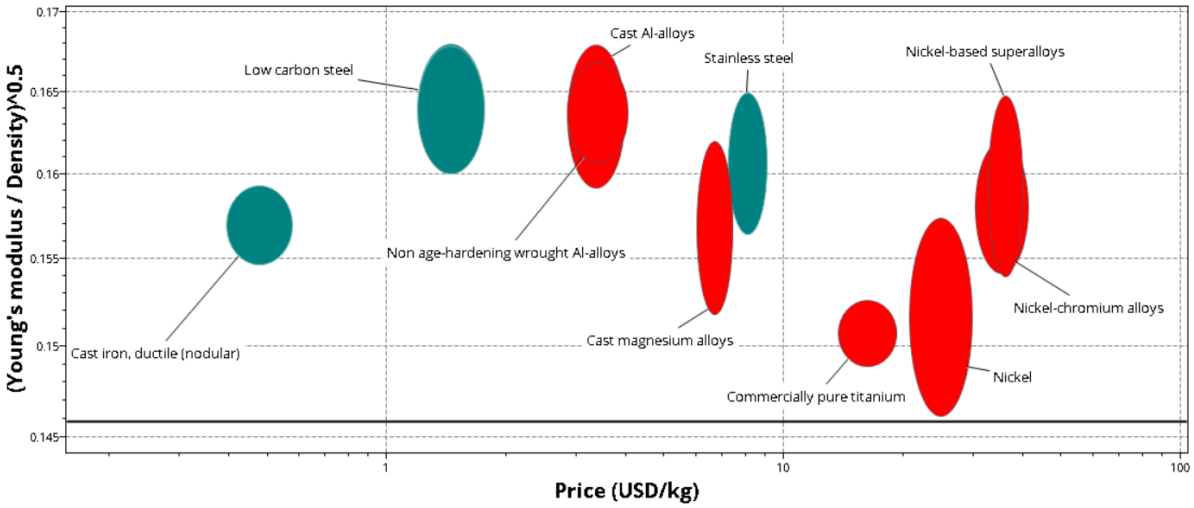


Figure 6: Top 11 materials can be choose for designing blade of a turbine engine

Table 1: Decision matrix based on 4 criteria: high natural frequency, low price, high Young's modulus and low density

Material	Natural Frequency $(\sqrt{\frac{MPa}{kg/m^3}})$	Price (USD/kg)	Young's Modulus (GPa)	Density $(1000\ kg/m^3)$
Cast iron, ductile (nodular)	0.155 - 0.159	0.396 - 0.575	170 - 180	7.05 - 7.15
Low carbon steel	0.16 - 0.168	1.2 - 1.75	200 - 220	7.8 - 7.82
Medium carbon steel	0.16 - 0.168	1.2 - 1.75	200 - 220	7.8
Cast Al-alloys	0.159 - 0.168	2.85 - 3.95	69 - 76	2.65 - 2.77
Non age-hardening wrought Al-alloys	0.161 - 167	2.86 - 4.04	69 - 74	2.63 - 2.7
Cast magnesium alloys	0.152 - 0.162	6.06 - 7.41	42 - 47	1.75 - 1.87
Stainless steel	0.156 - 0.165	7.24 - 9.05	190 - 210	7.61 - 7.87
Commercially pure titanium	0.149 - 0.153	13.7 - 19.1	100 - 105	4.51
Nickel	0.146 - 0.157	20.7 - 29.6	190 - 220	8.83 - 8.95
Nickel-chromium alloys	0.154 - 0.165	30.5 - 41	200 - 220	8.3 - 8.5
Nickel-based superalloys	0.154 - 0.165	32.7 - 39.9	200 - 220	7.88 - 8.68

From the data above, ranking material from 0 to 10, based on highest natural frequency, lowest price, highest Young's modulus, and lowest density.The highest total rank is chosen based on individual criteria. Based on the below matrix, the low and medium carbon steel are the best choice of material for building the blade for turbine engine. There is a trade for higher price in return of lower density.

Table 2: Decision matrix based on 4 criterias, including natural frequency, price, Young's modulus and density

Material	Natural Frequency rank	Price rank	Young's Modulus rank	Density rank	Total rank
Cast iron, ductile (nodular)	5	10	4	6	25
Low carbon steel	8	9	10	3	30
Medium carbon steel	8	8	10	4	30
Cast Al-alloys	7	7	2	8	24
Non age-hardening wrought Al-alloys	10	6	1	9	26
Cast magnesium alloys	2	5	0	10	17
Stainless steel	6	4	5	5	20
Commercially pure titanium	1	3	3	7	14
Nickel	0	2	6	0	8
Nickel-chromium alloys	4	1	10	1	16
Nickel-based superalloys	3	0	10	2	15

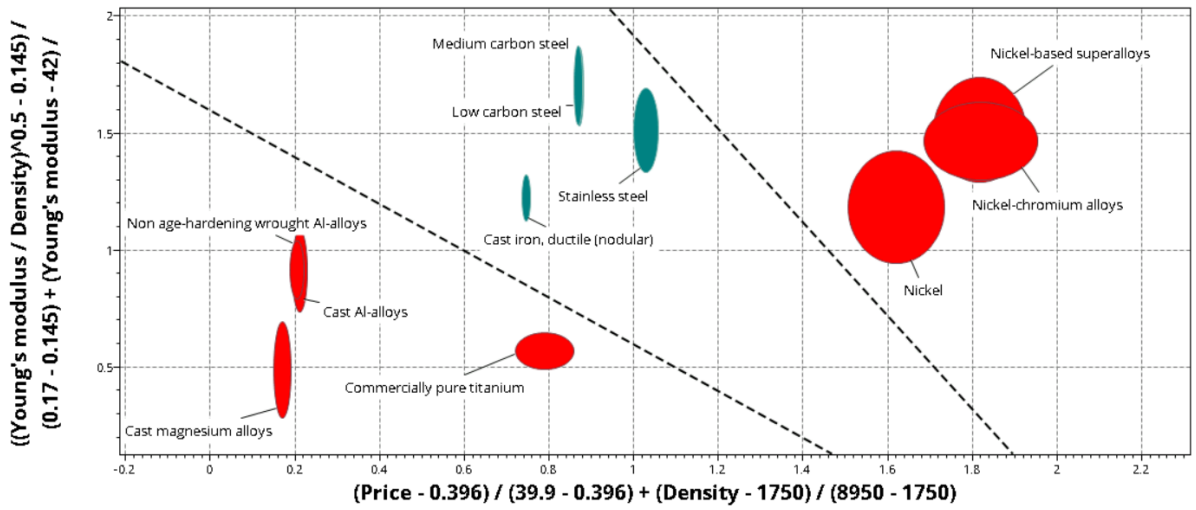


Figure 7: Visualizing the decision matrix with normalized data of each category*. Maximizing natural frequency and Young's modulus and minimizing price and density
*The data in the graph undergoes the equation for min-max feature scaling to make sure no criteria overwhelm one or another.

Combining the decision matrix table and the information from the graph, the top choice materials are medium and low carbon steel, alternative to stainless steel and ductile cast iron. This top tier choice exhibits a balance in all 4 criteria. The second tier choice satisfies the price and density conditions but

not the Young's modulus and natural frequency conditions, rank from non age-hardening wrought Al-alloys, cast Al-alloys, cast magnesium alloys to commercially pure titanium. The third tier meets the natural frequency and Young's modulus condition but fails the price and density condition, including nickel, nickel-chromium alloys and nickel-based superalloys.

Top three material for blade design are:

- Medium carbon steel
- Low carbon steel
- Non age-hardening wrought Al-alloys

Choosing through mechanical model

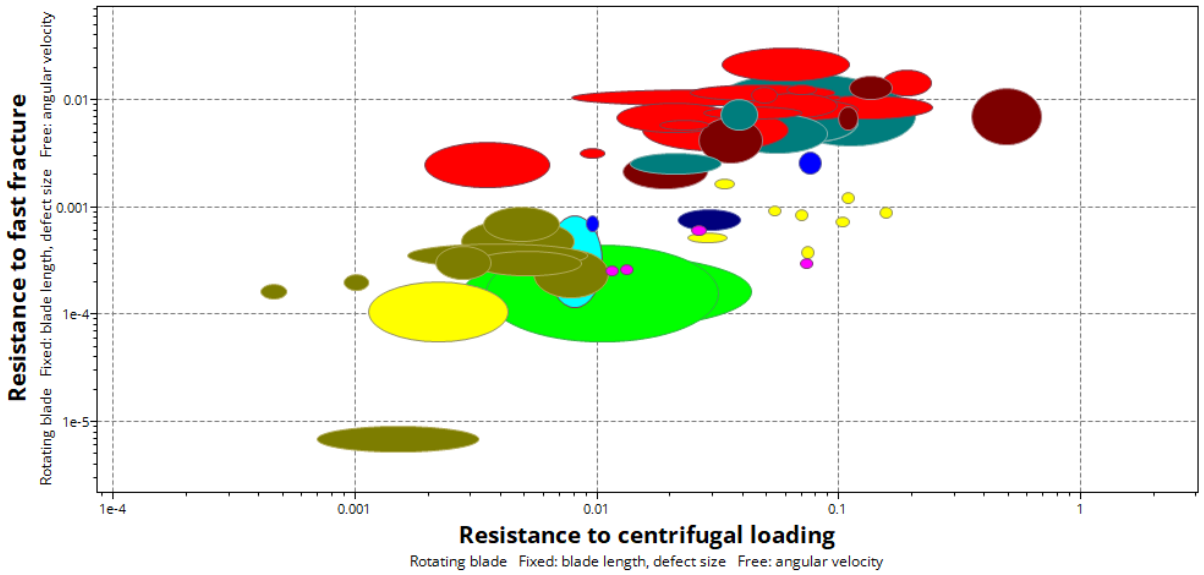


Figure 8: All materials in relations of resistance to fast fracture and resistance to centrifugal loading under rotating blade loading condition

In the configuration of the rotating blade, by default, the free variable is angular velocity; fixed variable is blade strength and defect size; limiting constraint is strength. The best way to maximize safety factor is to combine indices for "resistance to fast fracture" and "resistance to centrifugal loading".

- Equation for maximizing resistance to fast fracture: K_{IC}/ρ (The ratio of fracture toughness over density)
- Equation for maximizing resistance to centrifugal loading: σ_y/ρ (The ratio of yield strength, or elastic limit, over density)

There are 55 out of 100 materials that pass this stage, which further narrows the range of material when only choosing metals and alloys. Metal has relatively high elastic modulus, so it cannot deflect much under a load. They are generally ductile, which makes the material undergo a deformation process before rupture (more warning before failure for engineers). Most metals fail by ductile rupture after plastic deformation and "necking", which is a good thing for engineers.

The next step is to apply the fracture tolerance using the same equation and boundary as in Figure 2. The facture tolerance strength ratio limits the materials down to 28.

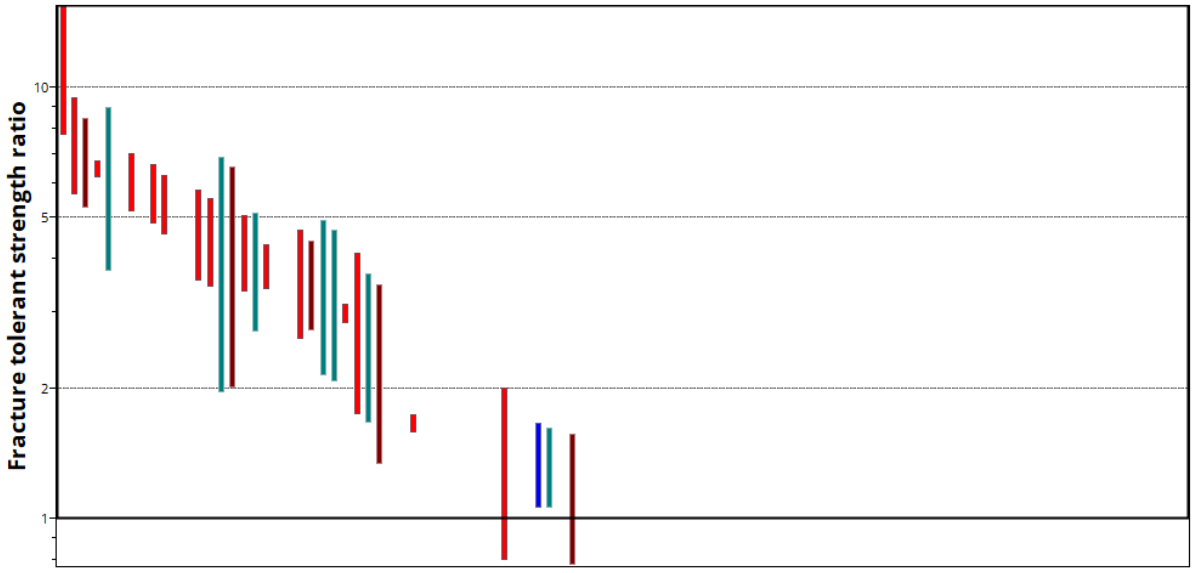


Figure 9: All material that passes the minimum fracture toughness over the scale of stress in relation to density

As the material is also subjected to bending stiffness or flexural modulus, the material is required to have high bending stiffness to resist the fast fracture condition. In general, the modulus in flexure is similar to the modulus in tension and compression and can be interchanged for approximate work. From choosing material according to the fatigue cycle equation, the rest of the steps lead to the same results of 11 materials for choosing the best material with additional parameters of choosing high resistance to fast fracture and resistance to centrifugal loading. Since these 2 parameters are more important overall, the individual is the additional of twice per rank into the final rank.

Table 3: Additional parameters into decision matrix

Material	Resistance to fast fracture	Rank	Resistance to centrifugal loading	Rank	Total rank	Final rank
Cast iron, ductile (nodular)	0.00328 - 0.00707	0	0.0346 - 0.0887	5	25	35
Low carbon steel	0.00533 - 0.0101	2	0.0327 - 0.0455	2	30	38
Medium carbon steel	0.0041 - 0.00922	1	0.0482 - 0.119	9	30	50
Cast Al-alloys	0.00701 - 0.0114	4	0.0435 - 0.0971	8	24	48
Non age-hardening wrought Al-alloys	0.0101 - 0.0139	8	0.0244 - 0.0946	1	26	44
Cast magnesium alloys	0.00663 - 0.00998	3	0.0287 - 0.119	6	17	35
Stainless steel	0.00736 - 0.0177	5	0.0332 - 0.147	4	20	38
Commercially pure titanium	0.0122 - 0.0133	9	0.0612 - 0.0798	10	14	52

Nickel	0.009 - 0.0124	6	0.00787 - 0.101	0	8	20
Nickel-chromium alloys	0.00952 - 0.0131	7	0.0434 - 0.0548	7	16	44
Nickel-based superalloys	0.0153 - 0.0304	10	0.033 - 0.109	3	15	41

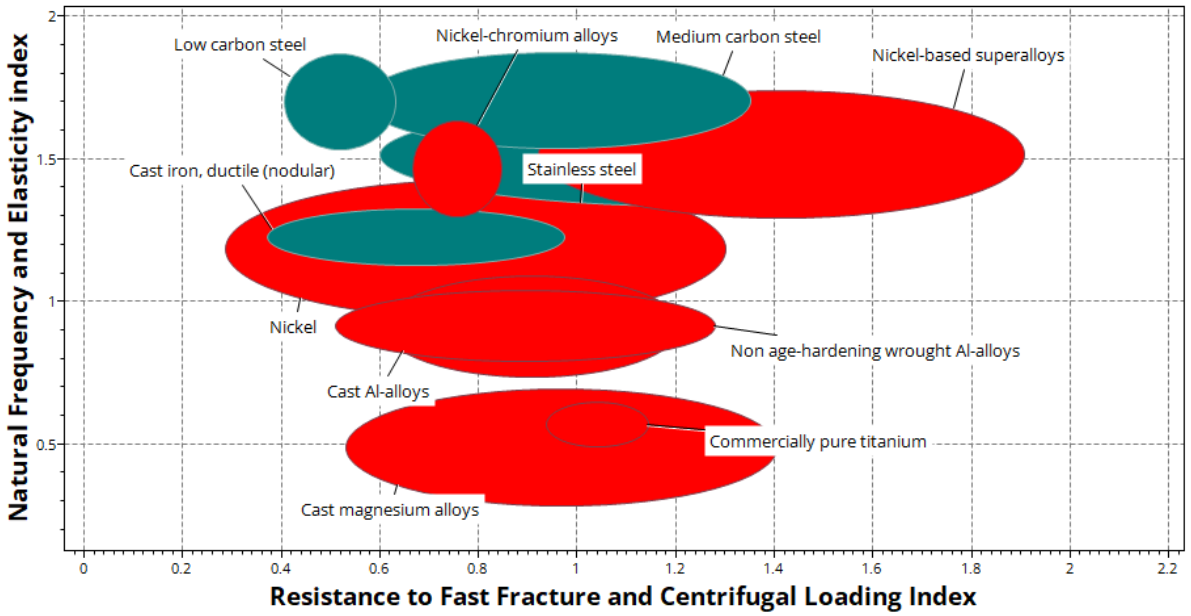


Figure 10: Comparison of materials between natural frequency, elasticity, and resistance to fast fracture and centrifugal loading conditions

Top materials are nickel-based superalloys, medium carbon steel and stainless steel for high index value in all criterias: natural frequency, elasticity, and resistance to fast fracture and centrifugal loading conditions. When price is factored in, the top choice material is medium carbon steel, stainless steel and nickel-based superalloys

Final material choice for mechanical design

1. Medium carbon steel
2. Stainless steel
3. Nickel-based superalloys

Eco audit

Then for the transportation we have selected 3 forms of transportation for the materials to travel from Beijing to Vancouver with rough estimates of distances between each point. The first distance is from Beijing to Tianjin, and this is traveled by a diesel train with a distance of 187000 meters. The second distance is overseas from Tianjin to Vancouver traveled with a Bulk Carrier, with a distance of 8,400,000 meters. The third distance is from Vancouver to a factory via semi-truck, with a distance of 40,000 meters. The masses used for each material was calculated with the density function; values of each material is shown in the chart below.

Number of blades	1000
End of Life	Landfill
Product Life	3 Years
Usage	250 Days/Year
Travelled Distance	10,000,000 m/Day

The energy consumption happens mostly during the operational phase, showing a significant performance in the long run but also expressing the need to minimise weight and ensure durability to avoid frequent replacements. Lighter and more durable properties are needed. Top choice of material for this purpose is medium carbon steel, stainless steel, then nickel-based superalloys.

Table 4: Overview of energy and climate change impact of all material

Phase	**Energy (MJ)**	**Energy (%)**	**Climate change (CO2-eq) (kg)**	**Climate change (CO2-eq) (%)**
Material	1.79e+07	0.4	1.01e+06	0.3
Manufacture	1.3e+06	0.0	1.01e+05	0.0
Transport	5.98e+07	1.3	4.01e+06	1.2
Use	4.49e+09	98.3	3.23e+08	98.4
Disposal	3.22e+04	0.0	2.25e+03	0.0
Total (for first life)	**4.56e+09**	**100**	**3.28e+08**	**100**
End of life potential	-6.63e+06		-3.18e+05	

Nickel-based superalloy contributes the highest CO2 emissions despite the manufacturing methods due to their energy-intensive production process. Meanwhile, medium carbon steel contributes the least amount of energy in the manufacturing phase. Top choice of material after factor in energy consumption and climate impact during its lifetime is medium carbon steel, stainless steel, then nickel-based superalloys.

Table 5: Energy use of material and carbon emission equivalent during its lifetime

Material	**Recycled content* (%)**	**Part mass (kg)**	**Qty.**	**Total mass (kg)**	**Energy (MJ)**	**%**	**Climate change (CO2-eq) (kg)**	**%**

Nickel-based superalloys	Virgin (0%)	23	1000	2.3e+04	7.8e+06	43.6	4.1e+05	40.7
Medium carbon steel	Virgin (0%)	23	1000	2.3e+04	7.3e+05	4.1	5.4e+04	5.3
Stainless steel	Virgin (0%)	23	1000	2.3e+04	1.6e+06	8.8	1.3e+05	13.4
Nickel-based superalloys	Virgin (0%)	23	1000	2.3e+04	7.8e+06	43.6	4.1e+05	40.7
Total			**4000**	**9.2e+04**	**1.8e+07**	**100**	**1e+06**	**100**

During the manufacturing process, nickel-based superalloy consumes the most energy and releases the most carbon emission. Due to the advanced properties of nickel-based superalloys, there is a high demand in energy to perform complex production techniques. While that, medium carbon steel and stainless steel have significantly lower energy requirements for manufacturing, making them more eco-friendly choices. The stainless steel even though has about twice the energy consumption and carbon emission, it is a relatively eco-friendly choice to nickel-based superalloys.

Table 6: Breakdown on energy usage and carbon emission during manufacturing

Component	**Process**	**Amount processed**	**Energy (MJ)**	**%**	**Climate change (CO2-eq) (kg)**	**%**
Super Ni Blade	Casting	2.3e+04 kg	2.5e+05	19.1	1.9e+04	18.4
Med Carbon Blade	Forging	2.3e+04 kg	1.2e+05	9.2	9e+03	8.9
Stainless Steel Blade	Forging	2.3e+04 kg	1.2e+05	9.1	8.9e+03	8.8
Super Power Blade	Metal powder forming	2.3e+04 kg	8.1e+05	62.5	6.5e+04	64.0
Total			**1.3e+06**	**100**	**1e+05**	**100**

Materials like nickel-based superalloys and stainless steel are recyclable, which overall reduced life cycle impacts. The energy recovery of nickel-based superalloys are lower and the material is only downcycle. It is repurposed into lower quality items, which overall better off the environment and explain why the material has higher energy use and environmental impact during the end of life. Medium carbon steel and stainless steel, when landfilled, result in lower disposal-phase emissions compared to more exotic material like superalloys.

Table 7: Breakdown of material in disposal

Component	End of life option	Energy (MJ)	%	Climate change (CO2-eq) (kg)	%
Super Ni Blade	Downcycle	1.2e+04	35.7	8.1e+02	35.7
Med Carbon Blade	Landfill	4.6e+03	14.3	3.2e+02	14.3
Stainless Steel Blade	Landfill	4.6e+03	14.3	3.2e+02	14.3
Super Power Blade	Downcycle	1.2e+04	35.7	8.1e+02	35.7
Total		**3.2e+04**	**100**	**2.3e+03**	**100**

Medium Carbon Steel:

- Lowest energy demand in manufacturing and use-phase, lowest CO2 emissions, and manageable end-of-life impact.
- A highly recyclable material, reducing environmental harm.
- Eco-friendliest choice overall without compromising mechanical properties.

Stainless steel:

- Lightweight, which minimizes use-phase energy.
- Moderate manufacturing energy and emissions, with high recyclability.
- A good eco-friendly alternative for applications where weight is a critical factor.

Nickel-Based Superalloys:

- Poor eco-friendliness due to high manufacturing energy and emissions, and only modest recyclability benefits.
- Best avoided unless exceptional high-temperature performance or strength is required.

Justification of material choice

Create penalty function

The penalty function is the sum product of normalized data and its weight to the final decision making function. The data is normalized according to the min max feature scaling function to make sure the data is always between 0 and 1. Weights are calculated based on the ratio of the individual rank to the total rank in the system of the equation, to make the final result of the penalty function always between 0 and 1. For this setup, every parameter is set to optimize its own to gain the global optimization.

``` Equations for penalty function
0.19047619047619* (([Young's modulus] / [Density])^0.5 - 0.145) / (0.17 - 0.145)
+ 0.166666666666667* ([Young's modulus] - 42) / (220 - 42)
```

+ 0.142857142857143* (1 - ([Price] - 0.396) / (39.9 - 0.396))
+ 0.0238095238095238* (1 - ([Density] - 1750) / (8950 - 1750))
+ 0.238095238095238* ([Fracture toughness] / [Density] - 0.003) / (0.0304 - 0.003)
+ 0.238095238095238* ([Yield strength (elastic limit)] / [Density] - 0.007) / (0.1 - 0.007)
```

Table 8: Parameters for decision making

| Parameter | Rank | Weight |
|---|---|---|
| Natural frequency | 8 | 0.1905 |
| Price | 6 | 0.1429 |
| Young modulus | 7 | 0.1667 |
| Density | 1 | 0.0238 |
| Resistance to fast fracture | 10 | 0.2381 |
| Resistance to centrifugal loading | 10 | 0.2381 |




Figure 11: Top three choice of material for mechanical design

# Calculation of manufacturing cost

Taken data from figure 1, the outer surface area is $0.588\,m^2$ and the inner surface are is $0.530304\,m^2$. The manufacturing cost is $6000\,CAD/m^2$, which makes the total cost is $6709.824\,CAD$ plus the cost of raw material, which will be determined in the next part of the report.

# Analysis from different viewpoints

Taking reference from the Eco Audit section, the nickel-based superalloys have the worst environmental impact, plus the price of raw material and manufacturing process are also high. The main reason is that the material has advanced properties that require energy-intensive manufacturing techniques to fabricate. While both medium carbon steel and stainless steel require less energy, so that they release less CO2 overall.
```

From the mechanical design viewpoint, the nickel-based superalloy has more superior properties compared to stainless steel and low alloy steel. The top material choice for mechanical design is made as nickel-based superalloys, medium carbon steel, then stainless steel.

From the accounting viewpoint, the fabrication price is constant since the dimensions are maximized at the beginning. The calculated fabrication cost is 6710 CAD plus the price of importing raw material. The price of raw material is what drives the material ranking. Top material choice for cost is medium carbon steel, stainless steel then nickel-based superalloys.

Overall, for a cheaper design and easier to manufacture, medium carbon steel is the best material to use. For more sophisticated design with high end jet engines, nickel-based superalloys can be used and considerations for material depreciation is required. Stainless steel is the close alternative to medium carbon steel for both mechanical design and manufacturing techniques.

Conclusion

The study evaluates materials for wind turbine blades across mechanical, environmental, and economic dimensions. Medium carbon steel is identified as the most suitable material for applications requiring cost-efficiency and eco-friendliness, thanks to its low manufacturing energy, recyclability, and robust mechanical properties. Stainless steel offers a good compromise between performance and cost but has a slightly higher environmental impact. Nickel-based superalloys, while mechanically superior, are limited by their energy-intensive manufacturing and significant carbon footprint, making them suitable for high-performance, high-cost applications. The penalty function approach effectively prioritizes decision-making criteria, providing a flexible and adaptable tool for optimizing material selection in diverse engineering contexts.

Recommendations

1. Improved Mechanical Analysis

- **Incorporate Dynamic Stress Simulations:** Introduce real-world simulations of dynamic loads, such as aerodynamic forces and wind gusts, to assess material behavior under varying operational conditions. This would provide a more comprehensive understanding of fatigue and vibrational performance.
- **Consider Surface Coating and Protection:** Evaluate the effects of surface treatments or coatings to improve impact resistance, reduce wear, and enhance environmental durability.
- **Aerodynamic Considerations:** Although simplified geometric shapes are used for material evaluation, incorporating aerodynamic analysis would provide insights into how material properties interact with blade shape efficiency.

2. Expanded Environmental Impact Assessment

- **Account for Recycling and End-of-Life Scenarios:** While the eco audit considers recyclability, it could be expanded to include detailed recycling potential (downcycling vs. upcycling) and recovery efficiency for each material.

- **Transportation Analysis:** Explore alternative transport routes or methods (e.g., green logistics) to reduce emissions during material delivery. Include variability in shipping distances to account for diverse sourcing scenarios.
- **Life Cycle Analysis Beyond Fixed Lifespan:** Extend the analysis to account for variable lifespans and maintenance cycles, reflecting realistic turbine operational durations.

3. Economic Factors

- **Market Price Volatility:** Consider material price trends and the impact of global market fluctuations, particularly for materials like nickel-based superalloys that are influenced by geopolitical and economic factors.
- **Life Cycle Cost Analysis:** Incorporate maintenance, repair, and replacement costs into the overall economic evaluation to provide a more holistic understanding of material affordability.

4. Enhanced Decision-Making Framework

- **Sensitivity Analysis of Penalty Function:** Conduct sensitivity analysis on the weights used in the penalty function to evaluate how variations in importance affect material rankings.
- **Incorporate Multi-Objective Optimization:** Implement optimization algorithms to identify materials that balance conflicting priorities, such as cost, performance, and sustainability, in a mathematically rigorous way.
- **Dynamic Weighting for Specific Applications:** Allow for application-specific adjustments in weight allocation (e.g., prioritize cost for low-budget projects or mechanical performance for high-performance designs).

5. Broader Material Considerations

- **Alternative Materials:** Explore newer material technologies such as advanced composites, carbon fiber, or bio-based polymers, which might offer better performance-to-weight ratios or sustainability benefits.
- **Hybrid Materials:** Consider combinations of materials (e.g., metal-composite hybrids) to optimize specific properties while reducing weight and cost.

6. Documentation and Presentation

- **Detailed Justification for Weight Assignments:** Provide explicit reasoning for the assignment of weights in the penalty function, enhancing transparency and reproducibility of the methodology.
- **Graphs and Visualizations:** Expand the use of graphical representations, such as Pareto frontiers or trade-off diagrams, to visually present the balance between different parameters.
- **Include Error Margins:** Discuss potential uncertainties or variances in the data (e.g., material property ranges) and their potential impact on the conclusions.